

Arnav Revankar

Junior Computer Engineer @ NJIT

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SUMMARY

Motivated Computer Engineering student with a passion for Embedded Systems. Experienced with STM32, ESP32, and RISC-V hardware/software development. Seeking Fall 2026 Co-op or Spring 2027 full time roles.

EDUCATION

New Jersey Institute of Technology (Sept 2023 - Present) Cumulative GPA: 3.65
BS Computer Engineering, Minor in Data Analytics Newark, NJ

- **Coursework:** Circuits and Systems 1 & 2, Microprocessors, Computer Architecture, Applied Machine Learning, Digital Design, and others.
- **Activities:** IEEE-HKN Inductee, 2026 Hackathon Volunteer, NJIT Solar Car Contributor (Hardware + Software Team)

TECHNICAL EXPERIENCE

NJIT Solar Car Member and Contributor May 2026 - Present

- Co-developed a 4 layer PCB for interfacing the NJIT Solar Car's peripherals (lights, indicators, etc) via the CAN Bus.
- Co-developing firmware for the Solar Car's STM32F413-based central control unit. Utilizing FreeRTOS for multithreading, and the STM HAL for interfacing peripherals.

Software Engineering Intern March 2025 - August 2025
Mathnasium Chatham, NJ

- Developed an email-based report generation and delivery software for a Mathnasium tutoring center.
- Currently being used to send 110-150 emails per day, saving 5 centers 30-40 hours a week.
- Effectively employs a BNF Grammar, that is used to generate syntactically valid text from customizable templates.

SKILLS

- **Languages:** C, C++, MATLAB, Java, Python, Go, SQL (MySQL), GNU Octave, L^AT_EX, Javascript, VHDL
- **Software Tools:** Linux/Unix, KiCAD, FreeCAD, STM32CubeMX & IDE, PlatformIO, ESP-IDF, Git, Docker, Excel
- **Build Tools/Debuggers:** Apache Maven, Apache Ant, CMake, Make, MSVC & GCC, GDB, Valgrind
- **Libraries:** STM32 HAL, FreeRTOS, NumPy, PyTorch, Scikit Learn, Matplotlib, Gnuplot
- **Communication Protocols:** UART, I2C, SPI, CAN, USB 2.0, TCP and Socket programming in C/C++ and Go

PROJECTS

8-Channel Logic Analyzer ([Github](#)) Feb 2026 - May 2026

- USB Logic Analyzer built around an STM32F446, capable of sniffing upto 500kHz signals.
- DMA data acquisition, and run-length encoded data transmission, to optimize byte usage on USB 2.0 FS.
- UART, SPI, I2C, and CAN signal decoding via a custom Go backend.

BitSum Lecture Summarizer ([GitHub](#)) Oct 2025 - Dec 2025

- Built a lecture summarizer tool, integrating Microsoft BitNet, OpenAI Whisper, and a custom feedforward network.
- Transcribes and summarizes lectures on CPU, 2x faster than human comprehension on average.
- Integrated with Docker to ensure reproducibility.

SolarVision Autonomous Mower ([Github](#)) Sept 2025 - May 2025

- YOLO and LiDAR-based autonomous lawnmower, engineering capstone project.
- Designed a custom 6-layer PCB for STM32N6 MCU, with external SPI NOR Flash and SDRAM Memory through the FMC.
- Trained a YOLOv8 Segmentation model on a custom dataset, and exported to C for the STM32N6 NPU, with the ST Edge AI Tool Suite.
- LiDAR data acquisition and path planning on an ESP32S3

ESP32 2nd Stage Bootloader May 2026 - Present

- Building a toy 2nd stage bootloader for an ESP32S3 microcontroller.
- Planning on supporting basic peripheral initialization, and OTA firmware updates if possible.
- Written in C using ESP-IDF HAL library and tooling, as well as Doxygen for code documentation.